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Multimodal oral cancer detection with embedding-level oversampling of fused image and clinical data

Pablo Ormeñ o-Arriagada[1,2,5,8]✉, Carla Taramasco[2,5,8], Carlos Toro[3,8], Gastó n Má rquez[2,4], David Araya[2,5,6,8], Valentina Z u n ˜ iga[5], Diego Mellado[2,5,8], Diego Alcocer[6,7,8] & David Ruete[5,8]

Oral cancer is a highly aggressive disease that is often detected too late, resulting in poor survival outcomes. Screening methods are limited by subjective assessment and by the scarcity of confirmed cancer cases in real-world datasets. We developed a multimodal approach that combines photographic features with basic clinical information to improve early identification of malignant lesions while addressing the strong imbalance typically found in medical datasets. Multiple machine learning models were trained on a dataset of annotated oral images paired with patient metadata, using synthetic sample generation techniques to improve representation of cancer cases. Our best configuration achieved good macro F1 and Area Under Receiver Operating Characteristic, outperforming image-only baselines. The improvements indicate that integrating contextual patient information was associated with improved class-balanced performance under the evaluated protocol. These findings reflect performance differences under the evaluated multimodal protocol and imbalance learning for assisting clinicians in earlier recognition of oral cancer. Our key contribution lies in embedding-level oversampling within a fused image–metadata space, which preserves latent semantics while improving minority-class sensitivity. These results arise from a tightly controlled, single-center experimental setup and are best understood as exploratory insights into the methodology, rather than as evidence that can be readily extended to broader clinical contexts. These findings reflect performance differences within the evaluated experimental setting. Metadata integration is treated as a structural component of multimodal representation within a constrained experimental framework.

Keywords Oral cancer detection, Multimodal learning, Class imbalance, SMOTE, ADASYN, XGBoost

Oral Cancer (OCA) represents a significant global health challenge, characterised by high incidence and mortality rates[1]. The World Health Organization reports that this disease causes approximately 200,000 deaths annually[2]. It is among the most prevalent head and neck malignancies worldwide, with late-stage detection strongly affecting prognosis and survival. Traditional diagnosis relies on visual inspection and biopsy confirmation, which are resource-intensive and prone to inter- observer variability[3]. This disease represents a significant global health burden, ranking among the most common cancers in low and middle income countries[4]. Early detection is critical because the 5-year survival rate drops drastically, from over 80% in early stages to less than 30% in advanced stages[5]. Despite being accessible to visual and tactile examination, oral cancers are frequently diagnosed late due to subtle early-stage symptoms and limited screening programs[6]. Early identification not only enhances survival but also reduces treatment-related morbidity and healthcare costs[7]. Integrating Artificial Intelligence (AI) driven diagnostic tools into clinical workflows has the potential to assist clinicians in identifying malignant

[1]Facultad de Ingenier´ıa, Ingenier´ıa Civil Informá tica, Negocios y Ciencias Agroambientales, Universidad Viñ a del Mar, Viñ a del Mar, Chile. [2]Centro para la Prevenció n y Control del Cancer, CECAN, Santiago, Chile. [3]Facultad de Ingenier´ıa, Universidad André s Bello, Talcahuano, Chile. [4]Departamento de Ciencias de la Computació n y Tecnolog´ıas de la Informació n, Universidad del B´ıo-B´ıo, Chillá n, Chile. [5]Facultad de Ingenier´ıa, Instituto de Tecnolog´ıa para la Innovació n en Salud y Bienestar, Universidad André s Bello, Viñ a del Mar, Chile. [6]Centro de Referencia de Salud, Hospital Provincia Cordillera, Santiago, Chile. [7]Unidad Emergencia Adultos, Hospital Dr. Só tero del R´ıo, Santiago, Chile. [8]Carla Taramasco, Carlos Toro, David Araya, Diego Mellado, Diego Alcocer and David Ruete have contributed equally to this work. ✉email: pablo.ormeno@uvm.cl

lesions earlier and 1 more accurately, thereby enhancing prognostic outcomes and guiding timely interventions[8]. Although oral cancers are accessible to visual inspection, imaging alone often fails to capture the full complexity of patient risk. Subtle clinical manifestations can be indistinguishable without contextual knowledge such as demographics, lifestyle factors, and anatomical site, motivating the integration of multimodal information. These challenges have prompted the exploration of artificial-intelligence-assisted diagnostics that can detect malignant patterns earlier and more consistently.

Multimodal data integration in medical diagnosis leverages the complementary strengths of diverse information sources to enhance discriminative diagnostic accuracy and robustness[9]. In the context of OCA detection, imaging data, such as intraoral photographs, cytology slides, or histopathology images, provide rich visual features that capture morphological changes associated with malignancy[10]. However, visual data alone may be insufficient to fully characterize patient-specific risk. Integrating patient metadata, such as age, sex, and clinical risk factors, provides additional contextual information that may support clinically informed decision-making; however, this integration does not in itself constitute model interpretability as defined in the explainable AI literature[11–13] . By combining these modalities, diagnostic models can learn richer feature representations that account for both visual and non-visual cues, leading to more nuanced and individualized predictions[14]. Recent advances in Deep Learning (DL) have enabled effective fusion strategies for multimodal biomedical data, paving the way for precision diagnostics in oral oncology[15]. However, multimodal datasets often heighten class imbalance. While abundant healthy or benign samples exist across modalities, malignant and premalignant cases remain limited, producing highly skewed distributions that compromise model sensitivity in exactly the cases of greatest clinical importance[16]. Recent multimodal cancer studies have achieved Area Under Curve (AUROC) improvements of 10–20 % and accuracies above 90 %, underscoring the superior diagnostic performance of integrated data approaches compared with unimodal models. Recent transformer-based multimodal systems have demonstrated strong performance in medical imaging tasks; however, such architectures typically require large-scale datasets and extensive computational resources. The present study intentionally restricts architectural complexity in order to isolate imbalance mitigation behaviour within fused embeddings under constrained data conditions [17]. Nevertheless, no study has systematically examined imbalance-aware learning within fused image and metadata embeddings for oral cancer detection. We address this gap by evaluating embedding-level imbalance-handling approaches within a multimodal fusion framework and analyzing their effect on class-balanced performance measures for oral cancer classification.

Machine Learning (ML) and DL have emerged as promising tools to assist in early detection through pattern recognition of oral lesions[18]. Their performance often suffers from the substantial class imbalance typical of medical datasets, where early-stage or malignant lesions represent only a small subset of cases. This imbalance is particularly problematic in diagnostic tasks, as positive disease instances are considerably rarer than healthy or benign samples, leading to reduced model sensitivity for clinically critical cases[19]. It also can bias model learning toward majority classes, reducing sensitivity for clinically critical cancer cases. In oral cancer datasets, for example, healthy and benign samples often outnumber malignant or pre-malignant cases by large margins[20]. Class imbalance in cancer datasets biases models toward majority classes, resulting in high accuracy but reduced sensitivity for detecting malignant and pre-malignant cases.[21]. Such misclassifications are unacceptable in clinical settings, where false negatives can delay diagnosis and worsen outcomes[22]. Moreover, imbalanced data can distort decision boundaries, hinder generalization, and reduce the reliability of performance metrics like accuracy. Addressing this issue requires specialized techniques such as data resampling, cost-sensitive learning, or hybrid augmentation strategies that preserve clinical validity while improving model fairness and sensitivity[23]. Previous research in oral cancer diagnosis has either analysed imaging and patient metadata separately or relied heavily on raw image augmentation to balance datasets[24,25]. Systematic evaluation of oversampling in fused multimodal embeddings remains limited, hindering balanced integration of visual and contextual data[26]. Consequently, imbalance-aware multimodal learning could generate representative minority samples and enhance sensitivity in early oral cancer detection[27]. Within this context, recent studies have explored multimodal learning frameworks for oral cancer detection by integrating imaging data with clinical or contextual information[28–30]. These works primarily focus on end-to-end system design and performance optimization across heterogeneous acquisition settings. In contrast, the present study investigates how embedding-level imbalance mitigation interacts with multimodal fusion under limited and imbalanced data conditions, providing complementary insight into model stability and minority-class sensitivity rather than proposing a new screening system.

To mitigate class imbalance in training datasets, several oversampling techniques have been developed to synthetically generate new samples for underrepresented classes[31]. Among the most widely adopted are Class Weighting (CW), Synthetic Minority Over-sampling Technique (SMOTE) and Adaptive Synthetic Sampling (ADASYN). SMOTE generates synthetic data by interpolating between existing minority class samples, effectively introducing plausible intermediate representations in the feature space without simply duplicating original instances[32]. This helps the model learn smoother class boundaries and reduces overfitting. ADASYN extends this idea by focusing the oversampling process on harder-to-learn minority samples—those near class boundaries, thereby adaptively shifting the decision boundary to improve minority class detection[33]. Both techniques are particularly well-suited for tabular data or embedding spaces, such as those formed by deep learning models, where they help balance datasets and enhance model sensitivity to rare but clinically important cases. Additionally, embedding-space.

oversampling preserves the latent semantic structure of multimodal representations, ensuring consistency between visual and contextual features. It also offers a computationally efficient alternative to pixel-level augmentation, reducing processing overhead while maintaining meaningful data variability.

This investigation aims to evaluate the directional improvement of multimodal fusion in addressing class imbalance in oral cancer classification[34]. Our approach proposes a multimodal learning approach that integrates image embeddings with patient metadata to improve early oral cancer detection. Combining complementary

information sources and handling data imbalance will enhance model reliability and clinical applicability. We hypothesise that applying oversampling within a fused image–metadata embedding space improves minority-class classification compared with unimodal or unbalanced baselines[35]. Specifically, we propose a fusion strategy that combines image embeddings, extracted from a pretrained Convolutional Neural Network (CNN), with structured patient metadata to form unified feature vectors. These fused representations are then used as input for oversampling techniques such as SMOTE and ADASYN, applied directly in the embedding space. The objective is twofold: first, to generate synthetic representations of minority classes that preserve both visual characteristics and associated clinical context; and second, to improve classifier performance, with particular emphasis on sensitivity to underrepresented conditions such as oral cancer. However, most existing studies focus solely on image data and overlook complementary clinical metadata such as age, sex, and risk factors. Persistent class imbalance in available datasets further limits model generalisation[36]. In summary, the study evaluates the combined effect of multimodal fusion and embedding-level imbalance mitigation on oral cancer classification under limited and imbalanced data conditions. By comparing image-only and multimodal configurations with and without oversampling, it provides evidence that balancing fused representations improves sensitivity to minority cancer cases. These findings support the use of embedding-level augmentation as a practical and reproducible strategy in multimodal oral oncology research. This study provides an empirical evaluation of embedding-level imbalance mitigation for oversampling within multimodal feature spaces, reducing reliance on direct image-level augmentation, together with empirical evidence that balancing fused embeddings enhances recall and macro F1 for minority oral cancer cases. In addition, a reproducible evaluation on real-world clinical data demonstrates the diagnostic relevance of embedding-level augmentation in AI-based oral oncology. By addressing data imbalance and the limited integration of patient metadata, the proposed approach improves sensitivity to rare malignant lesions and establishes a methodological foundation for the development of more equitable and generalisable multimodal models in oral cancer research. Accordingly, this work contributes a structured configuration comparison under constrained experimental conditions, examining how fusion and imbalance handling strategies influence class-balanced performance within a fixed experimental setup. Moreover, we examine the behaviour of embedding-level imbalance mitigation within a multimodal image–metadata classification framework under constrained data conditions, with the objective of evaluating whether latent-space resampling after fusion influences class-balanced performance within a fixed architectural configuration. In this context, the present study does not aim to establish state-of-the-art performance relative to contemporary transformer-based multimodal architectures, but instead isolates the behaviour of embedding-level resampling within a fixed and tractable architectural configuration. Indeed, direct benchmarking against large-scale transformer systems would require substantially larger datasets and fundamentally different experimental objectives, and therefore falls outside the scope of the current work.

The structure of the paper is as follows. In Section Related Work situates this study within the existing literature on oral cancer detection, multimodal learning, and imbalance mitigation strategies, highlighting current approaches and identifying key methodological gaps that motivate our work. In Section Methods, we present the dataset characteristics, preprocessing pipeline, and the multimodal fusion framework that integrates image embeddings with clinical metadata. Section Results reports the comparative performance of four classifiers under different imbalance-handling strategies, including CW, SMOTE, and ADASYN. Finally, Section Discussion interprets the findings in light of existing literature, examining the clinical and methodological implications of embedding-level oversampling for oral cancer detection.

Related work

Recent advances in medical image analysis have been largely driven by DL architectures, particularly CNN and hybrid attention-based models. Numerous studies have demonstrated high performance in cancer detection and segmentation tasks through increasingly sophisticated architectures and optimization strategies. For example, segmentation-focused approaches such as CICADA (UCX) achieve high Dice and AUC scores by integrating feature selection and optimization techniques into U-Net-based pipelines[37]. Similarly, hybrid DL models incorporating attention mechanisms and multi-scale feature extraction have been successfully applied to thyroid ultrasound, liver tumor, and prostate cancer segmentation, improving robustness against noise and variability in imaging conditions[38–40].

Beyond segmentation, classification-oriented frameworks have explored combinations of DL and optimization algorithms to enhance predictive performance. Hybrid architectures integrating feature selection and attention mechanisms, such as AKAttNet combined with electromagnetic interaction-based optimization, have demonstrated strong results in neurodevelopmental disorder classification tasks[41]. In oncology contexts, multimodal and hybrid learning strategies have also been explored for tasks such as.

drug response prediction, where heterogeneous data sources are integrated to improve predictive performance[42]. Additionally, attention-based and pyramidal architectures have been proposed to improve feature representation and classification accuracy across heterogeneous datasets, often achieving high performance through complex architectural designs[43,44].

In parallel, several works have explored the role of preprocessing, feature engineering, and model selection in improving classification outcomes in medical imaging. Comparative studies across architectures such as ResNet, DenseNet, and MobileNet highlight the importance of representation learning and optimization strategies in achieving high accuracy in tasks such as diabetic retinopathy detection[45]. Similarly, optimized hybrid frameworks for breast cancer classification demonstrate how feature selection and attention-based mechanisms can enhance predictive performance under varying imaging conditions[42]. Reviews of deep learning and transformer-based methods further emphasize the growing dominance of large-scale architectures, particularly Vision Transformers and hybrid models, which leverage extensive datasets and computational resources to achieve superior performance[46].

Explainability has also emerged as a relevant direction in medical AI, with approaches such as attention visualization and Grad-CAM being applied to improve interpretability in skin cancer detection models[47]. However, systematic reviews indicate that despite strong reported performance, a persistent gap remains between experimental results and real-world clinical applicability, often due to limited external validation, lack of interpretability, and insufficient evaluation under heterogeneous conditions[48].

With respect to class imbalance, most existing approaches rely on data-level or algorithm-level strategies, including oversampling methods such as SMOTE and ADASYN, cost-sensitive learning, and data augmentation techniques[49]. These methods are well established in structured and tabular domains and are typically applied prior to training or within the learning objective. However, their application in multimodal medical imaging—particularly in feature fusion settings—remains comparatively underexplored[50]. Existing studies generally address imbalance at the input or model level without explicitly analysing its interaction with fused latent representations.

Recent advances in biomedical AI increasingly emphasize explainability, collaborative validation, and clinical translation. Explainability frameworks, including SHAP and Grad-CAM, have been applied to improve transparency in diagnostic models, particularly in oncology imaging tasks[51]. In parallel, federated learning approaches enable collaborative model training across institutions without direct data sharing, addressing privacy and data governance constraints in clinical environments[52]. Clinical translation studies further highlight the importance of domain-shift robustness, regulatory compliance, and prospective validation when deploying AI systems in real-world healthcare settings[53].

While these developments are critical for translational deployment, the present study isolates imbalance mitigation behaviour within a constrained multimodal configuration. Additionally, the present work adopts a constrained experimental design to isolate the behaviour of embedding-level resampling within a multimodal image–metadata framework. Rather than proposing a novel algorithm or competing with large-scale architectures, this study provides a structured empirical examination of imbalance mitigation after feature fusion under limited data conditions[54]. This positioning allows for controlled analysis of imbalance mitigation effects without introducing confounding factors associated with large-scale architectural variation.

Methods

This section describes the methodological framework developed to improve oral cancer classification through Multimodal Data Fusion and advanced imbalance mitigation. We outline the dataset composition, preprocessing procedures, and feature extraction using pretrained CNN. Subsequently, we detail the integration of image embeddings with structured clinical metadata, the implementation of embedding-level oversampling techniques, and the evaluation protocols used to assess model performance. This framework operationalizes our hypothesis that embedding-level oversampling within a fused multimodal representation is intended to improve minority-class classification under the evaluated conditions. Three configurations were evaluated: (i) image-only classification models, serving as a unimodal baseline; (ii) multimodal fusion models combining image-derived embeddings with clinical metadata, without oversampling; and (iii) multimodal fusion models augmented with embedding-level oversampling to address class imbalance. This staged evaluation design enables a partial component impact analysis by isolating the effects of multimodal integration and oversampling strategies on classification performance. Quantitative differences across these configurations are reported using clinically relevant metrics, providing insight into the relative contribution of each component.

Proposed multimodal classification Pipeline

Figure 1 illustrates the overall workflow of the proposed multimodal oral cancer classification system. The pipeline begins with a preprocessing stage, where all images are standardised by resizing them to 224 × 224 pixels, adding a uniform border, applying a mild blur to reduce noise, and removing duplicates to ensure data quality.

In the feature extraction stage, high-dimensional embeddings (512 features) are generated from the preprocessed images using a CNN. These embeddings are then concatenated with structured patient metadata, including demographic attributes (age and sex) and behavioural risk factors (alcohol consumption, smoking, and betel-quid chewing), resulting in a multimodal feature representation.

The fused dataset is partitioned into a training set (70%) and a testing set (30%). The reuse of identical held-out test partitions across imbalance strategies was intentionally maintained to enable paired within-split comparison, while recognising that this design does not substitute for broader cross-validation. To address the inherent class imbalance, oversampling methods such as SMOTE and ADASYN, together with CW loss, are applied exclusively to the training data.

Multiple classifiers are trained on the rebalanced dataset, and model performance is evaluated on the unseen test set. Evaluation metrics include accuracy, precision, recall, macro F1, the area under the receiver operating characteristic (AUROC), and Cohen's Kappa, providing a comprehensive assessment of classification quality and scenario-specific improvement.

Finally, model selection in this study was guided by the need for architectural transparency, reproducibility, and feasibility under limited computational resources. Accordingly, established convolutional and multimodal fusion approaches were prioritized to enable controlled evaluation of fusion and oversampling effects.

The present analysis intentionally isolates embedding-level resampling after multimodal fusion to examine directional performance changes under a fixed architectural and training configuration. Alternative imbalance mitigation techniques, including weighted loss functions, focal loss, and image-level augmentation, were not included in order to preserve experimental control over the latent-space manipulation. This design does not imply methodological superiority of embedding-level resampling relative to these techniques, but rather isolates its behaviour within the defined protocol.

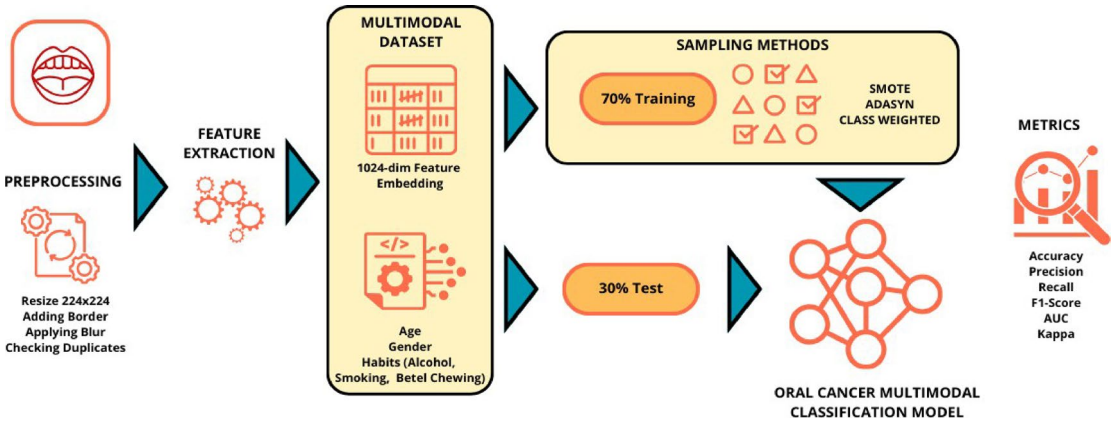


Fig. 1. Overview of the multimodal oral cancer classification pipeline. Images are preprocessed, embedded into 512-dimensional feature vectors, and fused with patient metadata (age, sex, lifestyle risk factors). The dataset is split into 70% training and 30% testing, with imbalance handled using SMOTE, ADASYN, or CW loss. Models are trained and evaluated using accuracy, precision, recall, macro F1, AUROC, and Cohen's Kappa. Preprocessing corresponds to Methods Proposed Multimodal Classification Pipeline, fusion to Methods Vector Fusion, and classification/evaluation to Methods Classifier Models; Evaluation.

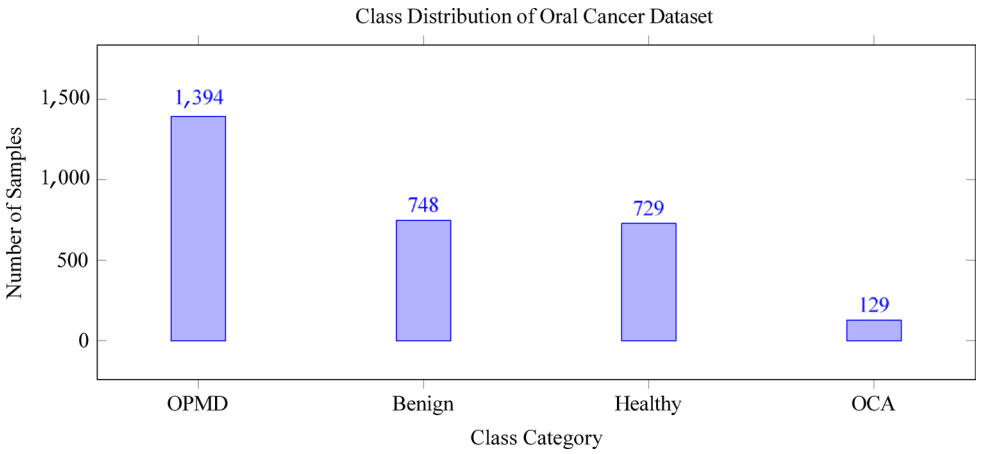


Fig. 2. Histogram showing class distribution in the oral cancer dataset. OPMD is the majority class, while OCA is the most underrepresented.

Dataset description

The dataset comprises 3,000 high-resolution images of oral cavities, captured using mobile phone cameras from patients within the Sri Lankan[55]. It includes documented patient and sample demographics, imaging modality information, and standardized annotation protocols, which are described here to support transparency and reproducibility, while acknowledging that the data originate from a single source and do not involve multi-center harmonization. Each image has been reviewed and categorised by domain experts into one of four clinical classes: Healthy, Benign lesions, Oral Potentially Malignant Disorders (OPMD), and OCA. Figure 2 illustrates the distribution of samples across the four diagnostic categories in the oral cancer dataset, with OPMD as the majority class and OCA cases represent only 4.3% of the dataset, underscoring the extreme imbalance.

In addition to the image data, each sample includes a structured metadata record that captures clinically relevant char- acteristics of the patient. These fields describe the *patient's age* at the time of diagnosis and *biological sex*, together with widely recognised behavioural risk factors linked to malignant transformation in oral tissues. Specifically, the metadata indicate whether the individual currently smokes or has a history of *smoking*, their *alcohol* consumption habits, and whether they *chew*

betel-quid, an established contributor to oral cancer in South Asian populations. The metadata also contain the expert-provided diagnostic label associated with the corresponding image. Together, these variables provide essential epidemiological context that complements the visual appearance of lesions and supports more accurate and clinically grounded classification.

Table 1 summarises the demographic characteristics and prevalence of key risk factors across the four diagnostic categories in the dataset. The statistics include sample counts, mean age, and proportions of male

patients, smokers, betel-quid chewers, and alcohol consumers, providing an overview of population distribution and risk exposure patterns relevant to oral cancer research.

We confirmed that the class counts reported in Figure 2 and Table 1 are internally consistent, with a total of 3,000 samples. We observed that the OCA group presented a higher mean age compared with other diagnostic categories; however, this observation is reported descriptively and not interpreted within a formal inferential framework.

In summary, the multimodal structure supports the integration of visual features with clinical metadata for early oral cancer detection, using a consistently annotated dataset drawn from a single institutional source[24]. The availability of aggregated patient metadata, multiple diagnostic classes, and transparent reporting of class distributions enables controlled evaluation of multimodal fusion and imbalance-aware learning strategies. At the same time, the single-source nature of the data is acknowledged as a limitation, providing a clear context for interpretation and motivating future validation on more diverse, multi-institutional datasets.

Feature extraction

We extracted high-level visual features from oral cavity images using a CNN based DenseNet121 architecture pretrained on ImageNet. Recent studies have demonstrated the strong performance of DenseNet121 in medical image classification tasks, particularly for oral and dermal lesion analysis[56]. All images were resized to 224×224 pixels and normalised to ensure consistent input distribution. The model was truncated before its classification head to preserve the deep convolutional representations learned during pretraining. DenseNet121 was chosen for its efficient feature propagation and strong ability to capture subtle patterns. Feature-level fusion with patient metadata incorporated relevant clinical covariates that may influence lesion characteristics.

The output of the final convolutional block was passed through a global average pooling layer, which reduced the spatial dimensions and produced a fixed-length image embedding vector of 512 dimensions for each image. These embeddings encode salient visual characteristics such as lesion shape, color intensity, texture, and anatomical context, making them well-suited for downstream classification tasks. The 512-dimensional projection was selected to balance representational richness and model generalisation[57].

The extracted image embeddings were subsequently concatenated with the corresponding tabular metadata vectors, comprising patient age, sex, and risk factor indicators, to form a joint multimodal feature vector. This fused representation, integrating both visual and clinical information, was then used as input to a fully connected classification head tasked with distinguishing between healthy tissue, benign conditions, OPMD, and confirmed OCA.

Vector fusion

Following feature extraction, we performed a late fusion strategy by concatenating the image embeddings with the corresponding patient metadata vectors. The image embedding vectors, obtained from the DenseNet121 architecture, had a dimensionality of 512 and captured rich spatial and semantic information from the oral cavity images. On the other hand, the metadata vectors were derived from structured clinical data, including age, sex, and lifestyle risk factors such as smoking, alcohol use, and betel-quid chewing. Metadata variables were incorporated as additional structured inputs within the fused embedding space. No formal attribution or feature-importance analysis was performed.

Prior to fusion, categorical metadata features were one-hot encoded, and continuous features were standardized using z-score normalization to ensure numerical compatibility across modalities. The resulting metadata vectors had a dimensionality of d_m, which was empirically determined based on the number of encoded features.

The two modalities were then integrated by performing a vector concatenation along the feature axis, yielding a unified multimodal vector of size $512 + d_m$. This fused representation was designed to capture both the morphological characteristics of the lesion and the associated patient-specific risk profile, enabling the model to make more informed predictions. The multimodal vectors were subsequently fed into deep learning classifier for final label prediction.

Data imbalance techniques

The dataset exhibited substantial class imbalance, with OCA cases markedly underrepresented compared with OPMD, benign lesions, and healthy controls. To enhance model sensitivity to minority cases, we applied oversampling and cost-sensitive learning within the multimodal feature space, which combined deep CNN embeddings with structured clinical metadata[21,58], improving gains in medical data models[59]. To correct this imbalance within the fused multimodal representation, we applied three complementary strategies. First, SMOTE generated additional oral cancer examples by interpolating between minority instances and their nearest neighbours in the embedding space, helping expand the decision region for malignant cases. Second,

Category	Count	Mean Age	Male (%)	Smoking (%)	Betel-Quid (%)	Alcohol (%)
Benign	748	46.9	58.8	10.0	15.0	4.8
Healthy	729	43.3	45.7	7.4	16.6	4.4
OCA	129	61.8	55.0	10.1	76.7	7.0
OPMD	1394	48.6	63.8	11.9	60.2	11.0

Table 1. Descriptive statistics by diagnostic category, including demographic and risk factor distributions.

ADASYN further concentrated new samples around difficult boundary areas, strengthening the model's ability to recognise subtle malignant patterns. Finally, a CW loss function increased the penalty for incorrectly predicting rare cases, ensuring they were prioritised during learning. Applying these methods directly to the fused feature vectors preserves the semantic structure of both visual and clinical information while improving minority-class sensitivity with minimal noise. Together, these imbalance-aware strategies promote more reliable discrimination of malignant lesions in a dataset dominated by benign and premalignant conditions .The strategy used to address class imbalance involved embedding-level oversampling applied after multimodal fusion within the embedding space, increasing minority-class representation without modifying the original images. Evaluation relied on macro-averaged metrics to ensure balanced assessment across classes, while alternative imbalance-mitigation techniques were not explored in order to maintain a controlled analysis.

Classifier models

We evaluated four complementary classifiers on fused multimodal feature vectors that combine 512-dimensional DenseNet121 image embeddings with preprocessed patient metadata. Logistic Regression (LR) offered an interpretable linear baseline with regularisation for stability in high-dimensional space. Support Vector Machines (SVM) with a radial basis kernel captured nonlinear boundaries driven by subtle embedding variations. Random Forest (RF) modelled heterogeneous feature interactions through tree ensembles that increase variance-controlled outcome to noise. XGBoost (XGB) further refined decision rules by sequentially boosting trees with explicit regularisation to control overfitting. This methodological diversity enabled a comprehensive assessment of multimodal learning performance and sensitivity to class imbalance in the context of early oral cancer detection. For transparency and reproducibility, full hyperparameter configurations for all classifiers are provided in the Supplementary Materials. Parameters were selected through iterative tuning based on validation performance and established best practices for high-dimensional biomedical data. All models were trained using a consistent optimization and training protocol across experimental configurations. A complete summary of hyperparameter values is provided in Table 5, provided in the supplementary section.

Evaluation

We evaluated the models using a stratified 70/30 train–test split repeated across four independent runs, each initialised with a distinct fixed random seed to support reproducibility and controlled estimation of performance variability. In each run, we applied resampling strategies, including class weighting, SMOTE, and ADASYN, exclusively to the training partition in order to prevent information leakage. We retained the same held-out test set across resampling strategies within each run to enable paired comparisons and consistent effect estimation. We report performance as mean ± standard deviation across runs, providing a reliable estimate of generalisation performance under limited and imbalanced data conditions. The evaluation protocol was deliberately designed to examine how model performance behaves under limited data, rather than to draw formal statistical conclusions. We relied on repeated stratified splits to track whether performance patterns remain consistent across runs, offering a practical view of directional stability. That said, this choice carries important limitations. It does not match the robustness of stratified k-fold cross-validation, nor is it intended for hypothesis testing. Instead, the approach should be seen as a pragmatic, exploratory first solution that balances methodological caution with the realities of working in data-constrained settings.

We assessed model performance using complementary metrics appropriate for imbalanced multi-class classification. Accuracy provided a global measure of correctness, while precision, recall, and macro F1 captured class-balanced behaviour and sensitivity to minority classes. We further quantified discriminative capability using the area under the ROC curve and accounted for chance agreement through Cohen's Kappa. To examine stable tendency, we analysed descriptive confidence

intervals across runs, which we interpret as exploratory indicators of performance stability rather than formal statistical significance, given the limited number of repetitions ($n = 4$). Together, these metrics provide a comprehensive evaluation of both overall performance and the ability to identify clinically relevant malignant cases in the presence of class imbalance.

In multi-class evaluation, both macro and weighted averaging were applied to summarise performance across classes. Macro averaging treats all classes equally, providing a balanced view of model performance, while weighted averaging accounts for class frequency, offering a more representative estimate under class imbalance. Combining both metrics ensures a comprehensive assessment of model behaviour across prevalent and minority cancer categories. These metrics collectively quantify both overall accuracy and minority-class sensitivity, directly addressing the study's central hypothesis. All experiments were implemented in Python using standard ML and DL libraries. Model training and evaluation were conducted on a workstation equipped with an AMD Ryzen 7 5800U processor, 16 GB of RAM, and solid-state storage, using integrated graphics rather than a dedicated GPU. The operating system, software framework versions, and hardware specifications are reported to support reproducibility. Random seeds were fixed where applicable to reduce nondeterministic variability across runs.

Reproducibility and resource availability

A public repository is provided at https://github.com/pormen/OCA_Embedding_Fusion, containing all resources required to reproduce the reported experiments, including data preprocessing scripts, multimodal fusion and embedding-level oversampling pipelines, model configuration files with fixed random seeds, and evaluation and metric computation scripts. These materials support exact replication of the experimental results and improve methodological transparency.

Results

This section presents the experimental results addressing the study's core hypothesis: whether embedding-level oversampling methods (SMOTE and ADASYN) within the fused image–metadata space improve classification performance—particularly macro recall and macro F1 for OCA, compared with raw fusion, CW training, and unimodal baselines. All models were trained and evaluated under a stratified 70/30 train–test split, repeated across four independent runs with fixed random seeds to ensure reproducibility. Resampling was applied exclusively to the training set in each run to prevent data leakage. The same held-out test set was reused across all imbalance-handling strategies and classifiers, enabling paired within-split comparisons. Performance metrics included accuracy, macro precision, macro recall, macro F1, AUROC, and Cohen's Kappa, providing a comprehensive evaluation of model repeatable pattern to class imbalance in a multiclass medical imaging context. Table 2 summarises the mean and standard deviation of test performance across runs for the four classifiers.

Ensemble-based classifiers exhibited performance differences when imbalance-handling strategies were applied. In particular, Random Forest and XGBoost showed increases in macro recall and macro F1 under resampling conditions relative to the raw configuration. Class weighting provided moderate changes, while resampling approaches, including ADASYN, were associated with higher sensitivity to minority classes within the evaluated protocol. These results reflect performance variation under the defined experimental configuration rather than comparative superiority across imbalance-handling strategies.

Class-wise analysis using XGBoost indicated that embedding-level resampling was associated with increased detection of malignant lesions relative to the raw configuration. Under the imbalanced setting, the model showed limited identification of OCA cases, whereas SMOTE and ADASYN were associated with increases in OCA F1. These changes occurred without substantial degradation in majority-class performance, with benign, healthy, and OPMD categories showing relatively stable macro F1 values across configurations. These observations reflect performance differences within the evaluated dataset rather than evidence of generalisable improvement.

Overall classification performance

Table 2 summarises the average performance of LR, RF, SVM, and XGB under different imbalance-handling strategies. Reported metrics include accuracy, macro-averaged precision, recall, macro F1, AUROC, and Cohen's Kappa, expressed as mean ± standard deviation across four runs.

The LR consistently produced the weakest results, indicating that linear models could not capture the complexity of fused image and clinical embeddings. Random Forest improved with resampling methods, particularly ADASYN, which raised its macro F1 compared with the raw condition. SMOTE also enhanced recall and agreement, highlighting the responsiveness of ensemble models to class imbalance. SVM achieved stable accuracy between 0.629 and 0.640, with minimal variation across strategies. Its highest macro F1 occurred under the Raw condition, suggesting limited benefit from resampling. XGB delivered the strongest overall performance. CW yielded the highest accuracy, while ADASYN and SMOTE improved recall and macro F1. XGB also achieved the highest AUROC values and Cohen's Kappa, confirming its bounded variability across folds under imbalanced conditions.

Overall, imbalance-handling strategies were associated with performance variation across classifiers, particularly for tree- based models such as Random Forest and XGBoost. Increases in macro recall and macro F1 were observed under resampling conditions relative to raw configurations. These findings should be interpreted as indicative of behaviour under the evaluated experimental protocol, without implying methodological superiority or generalisable advantage.

Strategy	Model	Accuracy	Macro Precision	Macro Recall	macro F1	AUROC	Cohen's Kappa
	LR	0.566 ± 0.018	0.486 ± 0.030	0.470 ± 0.029	0.475 ± 0.029	0.749 ± 0.018	0.340 ± 0.033
RAW	RF	0.594 ± 0.007	0.515 ± 0.018	0.394 ± 0.006	0.391 ± 0.008	0.806 ± 0.010	0.299 ± 0.012
	SVM	0.629 ± 0.016	0.683 ± 0.024	0.532 ± 0.027	0.550 ± 0.033	0.800 ± 0.008	0.437 ± 0.026
	XGB	0.649 ± 0.009	0.688 ± 0.044	0.488 ± 0.010	0.506 ± 0.015	0.828 ± 0.007	0.436 ± 0.014
	LR	0.557 ± 0.023	0.476 ± 0.037	0.479 ± 0.043	0.476 ± 0.040	0.749 ± 0.018	0.336 ± 0.038
CW	RF	0.589 ± 0.005	0.652 ± 0.150	0.389 ± 0.007	0.386 ± 0.011	0.813 ± 0.009	0.285 ± 0.011
	SVM	0.629 ± 0.016	0.683 ± 0.024	0.532 ± 0.027	0.550 ± 0.033	0.800 ± 0.008	0.437 ± 0.026
	XGB	**0.674 ± 0.017**	0.666 ± 0.036	0.551 ± 0.032	0.573 ± 0.038	**0.840 ± 0.010**	**0.493 ± 0.028**
	LR	0.551 ± 0.025	0.470 ± 0.034	0.472 ± 0.040	0.469 ± 0.037	0.741 ± 0.019	0.330 ± 0.041
SMOTE	RF	0.647 ± 0.015	0.666 ± 0.058	0.506 ± 0.032	0.535 ± 0.044	0.815 ± 0.003	0.433 ± 0.022
	SVM	0.640 ± 0.016	**0.725 ± 0.020**	0.499 ± 0.029	0.530 ± 0.038	0.806 ± 0.009	0.421 ± 0.029
	XGB	0.667 ± 0.014	0.624 ± 0.019	0.557 ± 0.033	0.578 ± 0.033	0.836 ± 0.010	0.481 ± 0.025
	LR	0.549 ± 0.032	0.467 ± 0.039	0.470 ± 0.043	0.466 ± 0.041	0.740 ± 0.020	0.327 ± 0.050
ADASYN	RF	0.652 ± 0.010	0.681 ± 0.043	0.526 ± 0.025	0.559 ± 0.034	0.816 ± 0.008	0.446 ± 0.014
	SVM	0.637 ± 0.017	0.718 ± 0.024	0.499 ± 0.029	0.528 ± 0.038	0.805 ± 0.009	0.419 ± 0.030
	XGB	0.665 ± 0.017	0.637 ± 0.008	**0.559 ± 0.040**	**0.580 ± 0.038**	0.835 ± 0.009	0.480 ± 0.030

Table 2. Average test performance (mean ± standard deviation across four runs with fixed seeds) under each imbalance-handling strategy. Resampling applied only to training partitions per run; identical held-out test splits enable paired comparisons.

Improvements in minority class detection

Detecting OCA cases is the primary clinical objective of this study. Models trained on raw data showed limited sensitivity (recall) to the minority class. LR and RF frequently misclassified OCA cases, and Support Vector Machine achieved only modest gains when imbalance remained unaddressed.

Imbalance-handling strategies were associated with increased minority class detection under the evaluated configuration, particularly in tree-based models. With ADASYN, RF recall increased from 0.394 to 0.526, and macro F1 rose from 0.391 to 0.559. XGBoost also benefited, achieving recall of 0.559 with ADASYN compared with 0.488 under raw data. These improvements were further reflected in AUROC and precision–recall curves, where resampling and CW shifted decision boundaries towards better identification of OCA cases while maintaining specificity for non-cancer classes.

XGB with CW or ADASYN achieved the best balance between sensitivity and overall accuracy. This trade-off is critical in clinical applications, where missing OCA cases poses greater risk than false positives. The results emphasise the importance of imbalance-aware learning for improving diagnostic sensitivity when using fused image and clinical embeddings.

Performance gains from multimodal integration

To evaluate the influence of different feature configurations, we trained a SVM classifier using stratified cross-validation across multiple input representations. Table 3 summarizes the results. The integration of embeddings with metadata achieved the highest accuracy, surpassing all unimodal settings. We evaluated model performance using features extracted through DenseNet121, decreased from 512 to 32, a gradual decline in performance was observed, indicating the importance of preserving richer feature representations. Notably, using metadata alone yielded the lowest accuracy, confirming that demographic and lifestyle attributes provide complementary but insufficient information in isolation. These findings highlight the performance gains achieved through multimodal fusion, where the combination of structured metadata with deep image embeddings enhances controlled variability and improves diagnostic reliability compared to unimodal baselines.The SVM baseline was implemented.

with an RBF kernel, probability estimates, $C = 1.0$, $\gamma =$ scale, and CW = balances. This model served as the reference to compare the impact of different feature configurations and imbalance handling strategies. All unimodal baselines used identical. Splits, seeds, and hyperparameter search ranges to ensure fair comparison against multimodal fusion.

The inclusion of both image embeddings and patient metadata was associated with improved model performance, confirming the complementary nature of visual and contextual information in oral cancer classification. The multimodal fusion achieved higher accuracy and greater stability across folds compared with embeddings alone, reflecting improved generalisation and quantified difference. This result supports the hypothesis that integrating heterogeneous data sources provides a more discriminative feature space, enabling more reliable differentiation between lesion types.

Exploratory stability and variance analysis

The robustness of the classifiers was examined through descriptive confidence interval analysis across four repeated train/test evaluations. Given the limited number of repeats ($n = 4$), these analyses are intended as exploratory stability assessments rather than formal statistical significance tests. Narrower intervals indicate greater consistency across runs, while upward shifts reflect improved sensitivity to minority cases. Table 4 reports 95% confidence intervals (CIs) for macro recall and macro F1, which are the most clinically relevant metrics in this context. Wider intervals reflect increased variability in model sensitivity to minority.

classes. Accordingly, the reported intervals should be interpreted as descriptive indicators of performance stability and effect direction.

Logistic Regression exhibited wide and overlapping intervals across all strategies, reinforcing its limited ability to capture minority class patterns consistently. Random Forest demonstrated a marked improvement in both recall and macro F1 under SMOTE and ADASYN, with intervals shifting upward relative to the Raw and CW baselines, suggesting that resampling improved reliability for minority class detection. Support Vector Machine maintained relatively stable intervals regardless of strategy, indicating robustness but little gain in minority class sensitivity. XGBoost produced the tightest and highest intervals under imbalance-aware methods, with ADASYN and SMOTE showing recall CIs extending above 0.55 and macro F1 CIs above 0.58, reflecting

Feature Configuration	CV Accuracy (mean ± std)
Embeddings + Metadata	0.576 ± 0.009
512 Embeddings	0.559 ± 0.028
256 Embeddings	0.556 ± 0.015
128 Embeddings	0.532 ± 0.017
64 Embeddings	0.521 ± 0.029
32 Embeddings	0.482 ± 0.038
Only Metadata	0.446 ± 0.023

Table 3. We compared cross-validation accuracy across different feature configurations derived from DenseNet121.

Model	Strategy	Macro Recall (95% CI)	Macro F1 (95% CI)
LR	Raw	[0.424, 0.516]	[0.429, 0.521]
LR	CW	[0.411, 0.547]	[0.412, 0.540]
LR	SMOTE	[0.408, 0.536]	[0.410, 0.528]
LR	ADASYN	[0.402, 0.538]	[0.401, 0.531]
RF	Raw	[0.384, 0.404]	[0.378, 0.404]
RF	CW	[0.378, 0.400]	[0.368, 0.404]
RF	SMOTE	[0.455, 0.557]	[0.465, 0.605]
RF	ADASYN	[0.486, 0.566]	[0.505, 0.613]
SVM	Raw	[0.489, 0.575]	[0.497, 0.603]
SVM	CW	[0.489, 0.575]	[0.497, 0.603]
SVM	SMOTE	[0.453, 0.545]	[0.470, 0.590]
SVM	ADASYN	[0.453, 0.545]	[0.468, 0.588]
XGB	Raw	[0.472, 0.504]	[0.482, 0.530]
XGB	CW	[0.500, 0.602]	[0.513, 0.633]
XGB	SMOTE	[0.504, 0.610]	[0.525, 0.631]
XGB	ADASYN	[0.495, 0.623]	[0.520, 0.640]

Table 4.. 95% confidence intervals for Macro Recall and Macro F1 across classifiers and imbalance-handling strategies. Values represent 95% confidence intervals across four folds.

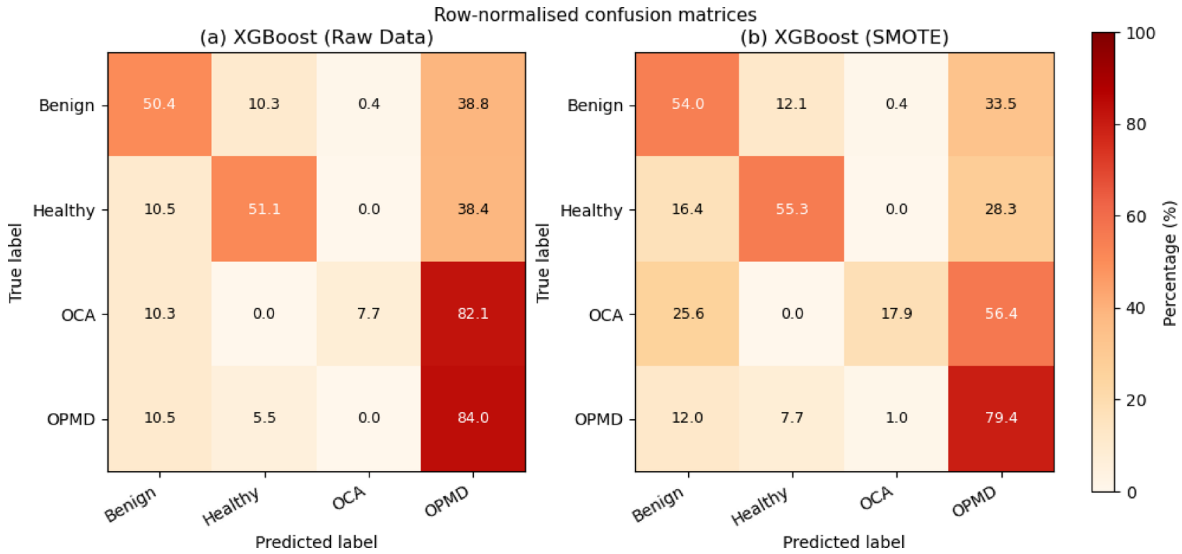


Fig. 3. Row-normalised confusion matrices for XGBoost models on the oral cancer dataset. Panel (**a**) shows results using raw data, while panel (**b**) presents results after applying SMOTE oversampling. Each row sums to 100%, allowing fair comparison of class-wise detection performance.

both improved performance and reduced variance across folds. Together, these findings confirm that ensemble methods, particularly XGBoost, achieve more reliable gains under resampling, whereas linear and margin-based models remain comparatively insensitive to imbalance-handling. The consistency of CI patterns supports the gains of the observed improvements and underscores the importance of combining ensemble learning with targeted imbalance mitigation.

Visualizations of key metrics

To further illustrate the impact of data balancing on classification outcomes, Fig. 3 presents the row-normalised confusion matrices obtained with the XGBoost models.

Figure 4 summarises the comparative performance of the classifiers under different imbalance-handling strategies, reporting the macro recall score.

Mutual information analysis confirmed that clinical metadata contributes independent discriminative signal, with betel-quid chewing and age showing the strongest associations with diagnostic category. This analysis quantifies statistical association between individual clinical variables and class labels but does not imply causal influence on classifier decisions within the fused latent space. This complementary information supports the

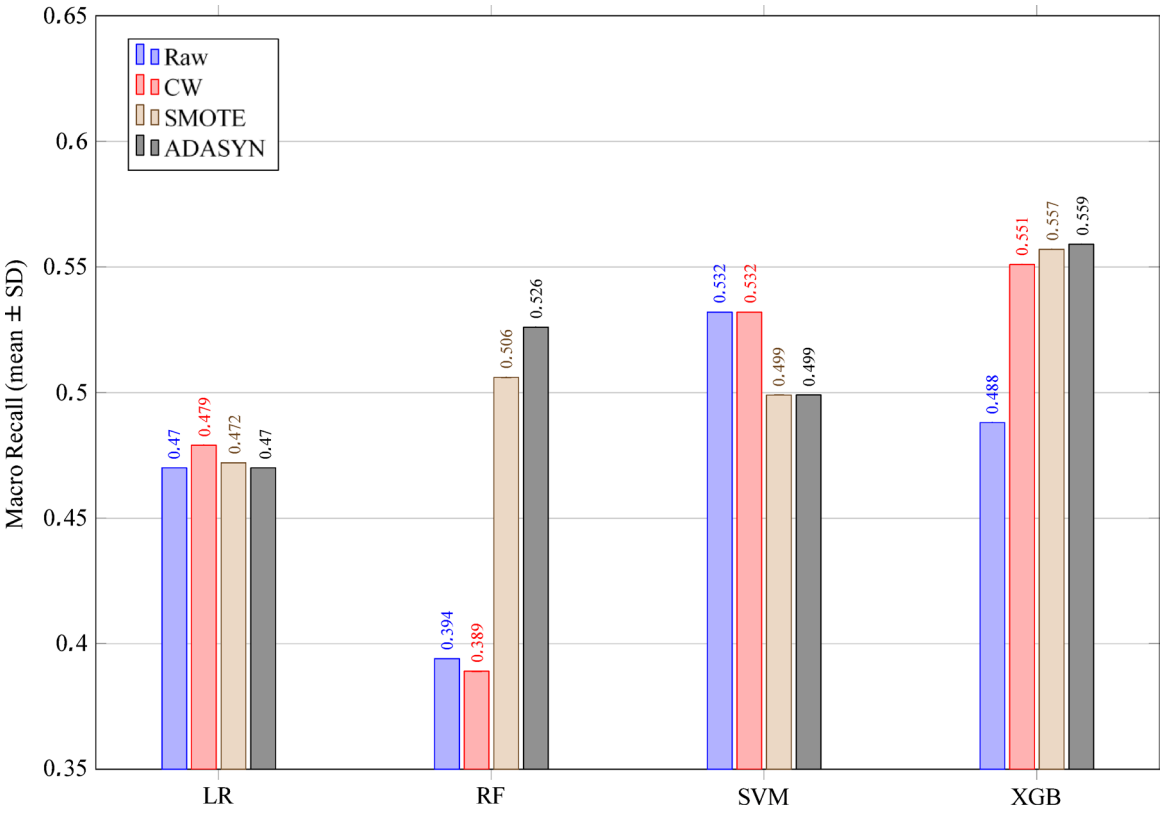


Fig. 4. Macro Recall (mean ± SD) for each classifier and imbalance-handling strategy. Error bars denote standard deviations across four runs.

performance gains observed in multimodal fusion by providing contextual cues that are not apparent from visual features alone, particularly for high-risk cases where early morphological changes may be subtle.

These results support the rationale for multimodal fusion. The top clinical predictors provide complementary information to the visual embeddings, which predominantly encode morphological patterns. In practice, integrating these metadata with image-derived features enables the model to condition visual cues on patient context (for example, risk behaviours or age strata), which is consistent with the observed gains in minority-class recall when using imbalance-aware strategies. The relatively modest MI for alcohol and smoking compared with chewing betel-quid suggests heterogeneity in how different behaviours manifest in the data, reinforcing the value of combining multiple clinical covariates rather than relying on any single factor.

Overall, this evaluation strategy provides a structured descriptive assessment of model behaviour under class-imbalance conditions within the defined experimental setup. Finally, granular error characterization, including confusion matrices, case-level qualitative inspection, and systematic false-negative analysis, was not performed in the present study. While such analyses are valuable in high-stakes diagnostic contexts, the current work prioritizes aggregate class-balanced performance metrics to evaluate imbalance mitigation under controlled conditions.

Discussion

This study takes a deliberately structured approach to examining how embedding-level imbalance mitigation behaves within a multimodal classification framework, particularly under the constraints of limited data. All findings are interpreted strictly within the boundaries of the experimental protocol. The focus remains firmly on methodological evaluation; it does not extend to formal explainability analyses, case-level diagnostic assessments, or clinically validated interpretations. Any reference to contextual metadata should therefore be understood as structural support within the model, not as evidence of causal influence or feature attribution. In that sense, the study is not intended to claim clinical readiness, inferential superiority, or broad generalisability. Rather, it offers a carefully controlled analysis based on a single-institution multimodal dataset, with conclusions that are explicitly limited to this specific experimental context.

The present study demonstrates that integrating multimodal learning with imbalance-aware strategies indicates performance differences in the detection of oral cancer from limited and heterogeneous datasets. By fusing image embeddings with structured clinical metadata and applying embedding-level oversampling, the proposed framework achieves consistent increases in recall and macro F1, particularly for minority cancer cases. The inclusion of metadata likely captures risk-factor correlations that complement morphological cues from images, aligning with known clinical patterns of oral carcinogenesis. To enhance clinical interpretability, future work will quantify variable-level contributions and report case-level explanations for positive OCA predictions.

In oral oncology, lesion morphology alone rarely provides a complete picture of disease risk or progression, as patient- specific factors such as age, sex, tobacco and alcohol use[60,61], and anatomical site significantly influence both the onset and biological behaviour of oral lesions. Integrating patient metadata allows the model to capture the interaction between tissue-level changes and external risk factors driving oral carcinogenesis. Tobacco and alcohol use, together with lesion location, strongly influence malignant transformation risk and histopathological severity. Embedding these clinical covariates alongside image features enables multimodal frameworks to better distinguish benign, premalignant, and malignant lesions while provides additional clinically relevant contextual information aligned with known risk factors. This approach provides a controlled methodological analysis by demonstrating that diagnostic accuracy depends not only on model depth but also on effective data representation and integration, ultimately supporting analysis under the evaluated experimental conditions.

Conceptual interpretation

The results show that combining patient metadata with image embeddings and applying imbalance-aware oversampling was associated with increased model sensitivity in oral cancer detection. Metadata integration enables the model to encode latent risk factors such as age, sex, tobacco use, and lesion site, complementing the morphological patterns captured by CNN-derived features. This enriched representation supports more discriminative decision boundaries when oversampling is applied. ADASYN improved macro recall by 0.13 and macro F1 by 0.17 in the Random Forest model, while XGBoost achieved consistent gains of about 0.07 across both metrics. These improvements enhance the model's capacity to detect minority oral cancer cases and reduce false negatives, reinforcing the potential of multimodal, imbalance-aware learning to support earlier and more accurate clinical diagnosis.

Comparison with prior work

Compared with previous image-based approaches, the proposed multimodal framework achieves a marked improvement in diagnostic performance. Earlier studies focused on image augmentation or unimodal pipelines[62], whereas the present embedding-level fusion integrates clinical metadata such as age, sex, and behavioural risk factors, including betel-quid chewing, smoking, and alcohol consumption—well-established contributors to oral carcinogenesis. Incorporating this contextual information enhances lesion discrimination and provides additional clinically relevant contextual information by aligning model decisions with recognised clinical and epidemiological risk patterns. The embedding-level oversampling strategy further strengthens performance by generating synthetic samples within a semantically consistent representation space, preserving meaningful clinical variability while mitigating class imbalance and redundancy. Ensemble classifiers, particularly XGBoost and Random Forest, achieved higher recall and macro F1, confirming their sensitivity gains over linear models for high-dimensional biomedical data with nonlinear dependencies[63,64]. Collectively, these results demonstrate that metadata-driven multimodal learning combined with embedding-level resampling enhances sensitivity and reliability in oral cancer detection. This aligns with recent evidence in head and neck oncology showing that integrating imaging with clinical and molecular data improves diagnostic and prognostic performance[65]. Moreover, systematic reviews support that oversampling techniques such as SMOTE improve minority-class sensitivity in medical datasets[21], underscoring the clinical and computational relevance of the proposed framework.

Clinical and computational implications

From a computational standpoint, the proposed embedding-level oversampling framework provides a controlled methodolog- ical examination of embedding-level resampling for balancing multimodal medical datasets. Also integrating multimodal embeddings with imbalance-aware strategies enhanced oral cancer detection, particularly for minority classes. Boosting-based models, especially XGBoost, achieved higher recall and macro F1 with SMOTE or ADASYN, confirming the advantage of ensemble learning for high-dimensional biomedical data. Unlike conventional image-space augmentation, which requires costly GPU operations and risks introducing unrealistic artefacts—synthetic resampling in the latent embedding space operates on compact vector representations, reducing training time and memory usage. This approach maintains semantic consistency while enhancing minority-class representation, enabling ensemble models such as XGBoost and Random Forest to learn more robust boundaries in imbalanced conditions. The resulting pipeline is computationally feasible within the experimental environment on standard hardware, particularly in resource-limited settings.

Recent work has demonstrated the importance of explainability in oral cancer screening systems, including approaches based on informed deep learning and case-based reasoning[12,66]. While such methods explicitly address interpretability and clinical transparency, the present study does not incorporate formal explainability mechanisms. Instead, it focuses on performance stability and imbalance mitigation, with explainability-oriented extensions positioned as an essential direction for future investigation.

Clinically, the improvement in minority-class sensitivity was associated with improved detection under the evaluated experimental conditions by reducing false negatives that often delay treatment and worsen outcomes. These gains are particularly relevant in low- and middle-income settings, where limited access to specialists and imaging resources constrains screening capacity. By combining photographic data with basic patient metadata, the proposed model supports potential AI-assisted triage applications, discussed as exploratory directions rather than validated deployment outcomes, aligning computational innovation with equitable healthcare delivery. Embedding-level oversampling offers a scalable solution for dataset rebalancing without the computational burden of image augmentation, facilitating potential applicability in resource-constrained settings, considered as a prospective direction rather than a validated deployment scenario. As a decision-support tool, the model supports analysis under the evaluated experimental conditions, prioritises high-risk lesions, and strengthens community screening. For researchers, the multimodal framework provides a reproducible foundation

for explainable and generalisable AI in oral oncology, advancing interpretability, domain adaptation, and longitudinal analysis of disease progression.

Clinical translation and regulatory considerations

Although the present study does not claim readiness for clinical deployment, several considerations would be required prior to translational application. External validation across independent institutions and imaging devices would be necessary to assess robustness under domain shift and demographic variability. Prospective evaluation within real-world clinical workflows would further be required to examine operational stability and diagnostic impact. From a regulatory perspective, AI-based diagnostic support systems must comply with evolving medical device frameworks, including performance monitoring, documentation of model updates, and post-market surveillance. Data governance, bias mitigation, and transparent reporting standards are essential to ensure ethical deployment. These requirements underscore that the current framework represents a methodological investigation under constrained experimental conditions rather than a clinically validated decision-support system.

Recent literature highlights the importance of federated validation, domain-shift robustness, and ethical auditing in biomedical AI deployment[49,50,54].

Emerging biomedical AI research increasingly incorporates explainability frameworks, federated validation strategies, and regulatory-aware evaluation pipelines. While such developments are critical for translational deployment, the present work isolates imbalance mitigation behaviour within a constrained multimodal configuration.

Limitations and future directions

The present study was conducted under intentionally constrained experimental conditions. These design choices define the scope of inference rather than reflecting methodological deficiency. All conclusions are bounded to the evaluated single-institution dataset and fixed experimental configuration.

Despite its promising results, this study presents several limitations that guide future research directions. First, the dataset size and single-center origin may constrain generalisability, and although repeated evaluation suggested consistent performance patterns, larger and more diverse cohorts are required to validate generalisability. Inter-observer variability in lesion annotation could also introduce bias in ground truth labels. To address this, future work will conduct multicentric validation involving patients across diverse geographic and clinical contexts, ensuring population and imaging heterogeneity representative of real-world screening conditions. We will also evaluate cross-site generalisation via external test sets and report domain-shift sensitivity (e.g., device type, lighting) using stratified analyses.

While metadata inclusion provided additional contextual information at the feature level, the model's internal decision logic remains opaque. We plan to integrate explainable AI tools such as SHapley Additive exPlanation (SHAP) values or integrated gradients to quantify the contribution of each variable, both visual and clinical, to model predictions, thus enhancing clinical trust and transparency.

Our architecture primarily relies on CNN-derived embeddings. Emerging transformer-based or attention-driven models could better capture contextual dependencies between lesion morphology and patient-level features. Comparative evaluation against multimodal transformers will help determine whether attention mechanisms further improve sensitivity and calibration. External validation and real-world lightweight clinical integration studies will assess the model's performance under clinical constraints such as variable lighting, device quality, and image noise. The ultimate objective is to evolve this framework into an explainable, low-cost, multimodal diagnostic assistant capable of reducing false negatives and accelerating early oral cancer. detection in community healthcare settings.

Although multimodal learning improved classification performance, several limitations remain. The dataset represents a narrow demographic and clinical scope, and annotation subjectivity or residual imbalance may affect generalisability. Broader multicentric validation is therefore required to confirm sensitivity gains. Even so, embedding-level multimodal integration was associated with improved detection of rare oral cancer cases within the evaluated dataset by combining morphological and contextual risk factors, a result of clear oncological relevance. The proposed framework provides structured analysis under constrained experimental condition by uniting computational accuracy with clinically meaningful interpretation, reinforcing its methodological relevance within constrained experimental settings in oral cancer detection.

The absence of detailed visual diagnostics such as confusion matrices or case-level error analyses represents a limitation of the current study. In particular, challenging or borderline cases may exhibit distinct error patterns that are not fully captured by aggregate metrics alone. Nonetheless, the use of macro-averaged measures and stability analysis partially mitigates this limitation by highlighting sensitivity to minority classes. A more granular exploration of error modes will be necessary to support clinical interpretability and trust.

Translating AI-based diagnostic systems into clinical practice requires careful consideration beyond predictive perfor- mance. Factors such as device heterogeneity, variability in image acquisition protocols, and population differences can substantially affect model generalization. Moreover, regulatory approval pathways, including compliance with medical device regulations and data governance standards, impose additional constraints on model design, validation, and reporting. Ethical considerations—such as transparency, bias mitigation, and the role of human oversight—are particularly critical in clinical decision-support contexts. The present study does not address these challenges directly, but instead provides a methodological foundation upon which such translational evaluations can be systematically built.

The use of macro-averaged metrics provides a more informative assessment of model behavior under class imbalance, particularly for rare or minority classes that are clinically critical. Performance gains observed under embedding-level oversampling suggest improved sensitivity (recall) to underrepresented cases, although some

variability remains due to limited sample availability. While weighted loss functions and data augmentation have shown promise in related studies, their exclusion in the present work allows clearer attribution of observed effects to the fusion and oversampling strategy. Future work will systematically compare these alternative imbalance-handling approaches under larger and more diverse datasets.

While recent transformer-based and hybrid architectures have demonstrated strong performance in medical imaging tasks, their evaluation often entails increased computational cost and reduced interpretability[67]. The present study instead emphasizes methodological clarity and controlled component analysis. Future benchmarking against such architectures will be essential to assess trade-offs between performance, efficiency, and clinical deployability.

Future work will extend the methodological analysis of the proposed framework through deeper interpretability, evaluation, and component-level assessment[68]. Dedicated explainability techniques, including visual attribution and feature-importance methods, will be integrated to better characterise the influence of image-derived embeddings and clinical metadata on model predictions, complemented by case-based analyses of correct and erroneous classifications. In parallel, richer evaluation protocols will be adopted, incorporating confusion matrices, threshold-based performance curves, and systematic error analysis to enable finer-grained inspection of minority-class behaviour. More comprehensive ablation studies will also be conducted to isolate the contribution of architectural components, preprocessing choices, and imbalance-mitigation strategies, supporting more robust attribution of observed performance gains.

In addition, future research will prioritise external validation across independent cohorts, imaging devices, and clinical settings to assess performance tendency and generalisability under real-world variability[69]. Multicentre and prospective studies will be required to evaluate clinical utility, workflow integration, and longitudinal performance in routine practice. These efforts will be complemented by investigations into alternative imbalance-handling approaches, computational efficiency, and resource-aware lightweight clinical integration, alongside consideration of data governance, ethical safeguards, and regulatory requirements. Together, these steps are essential for advancing the proposed approach toward clinically deployable, trustworthy, and responsible AI systems for oral cancer detection.

The absence of external validation, multicenter cohorts, and domain-shift testing precludes any claims of generalisability beyond the present dataset. Such evaluations represent necessary future research before translational consideration. . Similarly, the present findings should not be interpreted as evidence that latent-space synthesis is inherently superior to alternative imbalance mitigation strategies, but rather as an empirical observation within a fixed multimodal configuration. In addition, no formal latency benchmarking, memory profiling, or hardware comparison experiments were conducted. Therefore, no claims regarding suitability for mobile or resource-limited deployment can be made based on the present study. Computational observations are limited to feasibility within the experimental environment. Consequently, future investigations will incorporate feature-attribution methods such as SHAP and Grad-CAM, along with confusion matrices and structured false-negative analysis, to examine decision pathways and clinically relevant error patterns. Taken together, these limitations delineate clear directions for future investigation, including external validation while future research will prioritise external multicenter validation across heterogeneous imaging devices and demographic cohorts. Prospective studies designed to evaluate workflow integration, performance drift, and decision-support interaction will be required before translational consideration.

The novelty of this study does not lie in introducing a new imbalance mitigation algorithm. Instead, it lies in isolating and empirically examining the interaction between embedding-level resampling and multimodal fusion within a clinical image–metadata representation. While oversampling techniques such as SMOTE and ADASYN are well established in tabular domains, their behaviour after multimodal latent-space fusion has not been systematically characterised under constrained medical imaging conditions. The present work therefore contributes a structured empirical analysis rather than an algorithmic innovation. Accordingly, the findings should be interpreted as exploratory methodological observations within a single-institution experimental framework.

Data availability

The dataset analysed in this study is publicly available from Zenodo ${ }^{36}$. All code and configuration scripts necessary for replication will be provided upon publication.

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Author contributions

P.O.-A. led the study, designed the methodology, performed the experiments, analysed the data, and wrote the manuscript. C.T., G.M., D.A., V.Z., D.M., D.A and D.R contributed to study design, clinical interpretation, result validation, and manuscript review. All authors approved the final manuscript.

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Declarations

Competing Interests

The authors declare no competing interests.

Code and Reproducibility

Source code, configuration files, and scripts to reproduce splits, resampling, and analyses will be available in an open repository upon publication (link to be provided), enabling full computational reproducibility.

Use of Artificial Intelligence

Large language models were used to assist in language polishing and formatting. The authors are fully responsible for the content and interpretation of all results.

Ethics and Data Governance

This study used de-identified images and metadata from a public repository with appropriate usage permissions (see[55]). No protected health information was processed. We complied with dataset licence terms and institutional guidelines for secondary data analysis.

Additional information

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Correspondence and requests for materials should be addressed to P.O.-A.

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